



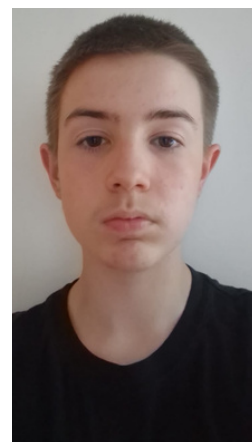
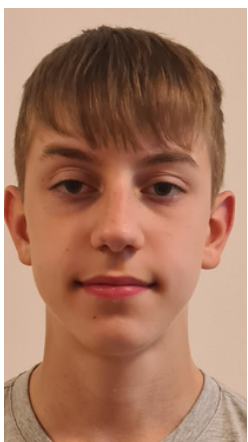
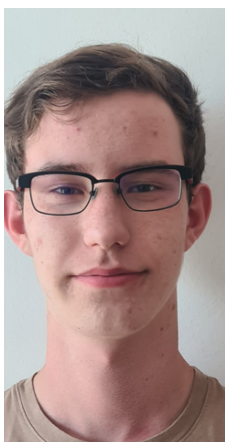
Robofreak Horizon

Category: Lightweight Entry



TEAM MEMBERS:

Matija Novak - team captain and programmer
Andrija Novak - robot constructor
Šimun Spevec - robot designer
Filip Podvezanec - hardware constructor
Ivan Kolarić - mentor



ABSTRACT

Our team consists of four team members (stated on the left). Our robot's body was fully desined and printed by us without any help from our mentor. As for the robot's software, we developed it using Arduino language with help of our mentor. So far we have been working on our robot since November of last year, making multiple hardware and software changes. All of us are relatively new to Soccer category, so we hope to learn something new that could help us improve in the future.

HARDWARE:

Our robot consists of two connected chassis and one smaller platform. The two chassis are connected via six metal pillars, while the platform is connected to the upper chassis using four plastic pillars. The connection between the two chassis is reinforced by motor mounts, which also hold the motors in place. The handle that we use to carry our robot is connected to the upper chassis with encased copper wires. All parts of the chassis, motor mounts, wheels, and handle were designed by us in Autodesk Fusion and 3D-printed with PLA plastic. Our robot has three double row wheels, and we use three Robotzone Planetary Gear Motors to rotate them. On our robot, there is also a motor driver for controlling the motor rotation, a microcontroller, an IR ball finder for detecting the ball, a bicolor display, a CAN Bus adapter, and a power supply. To power our robot, we use a LiPo battery. All those electrical components are placed on a platform. Furthermore, we added an ultrasonic distance sensor to have better field navigation. The last component we made for our robot is the dribbler. The dribbler consists of one motor that rotates the ball toward the robot.

SOFTWARE:

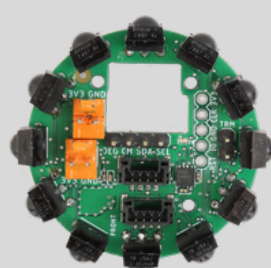
To create the program for our robot, we use the Arduino C++ programming language. The program starts when one of the buttons on the display is pressed. When an IR ball finder detects the ball, the robot starts to move towards the ball using trigonometric function:

$$Y = speed * \cos(rad - \pi/3) + rotation$$

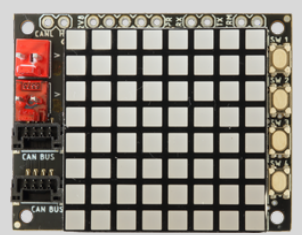
which depends on the ball's relative position to the robot. If the ball is in front of the robot, it moves forwar. However, if the ball is to the right or left, the robot moves in that direction while also moving a slightly backwards to get behind the ball. If the ball is position somewhere on the field where the robot cannot get behind it by just going forward, the robot moves sideways and backwards to achive this. If the ball is behind the robot, the robot moves backwards while also moving sideways in order to reach a position from which it can get behind the ball by going forward and sideways. When the robot reaches the ball, the dribbler starts rotating.



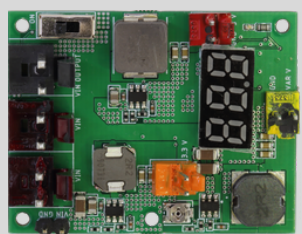
microcontroller -
mrm-esp32



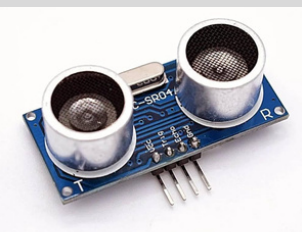
IR ball finder -
mrm-ir-finder3



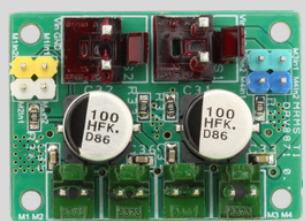
bicolor display -
mrm-8x8a



power supply -
mrm-ps3xe 0.1



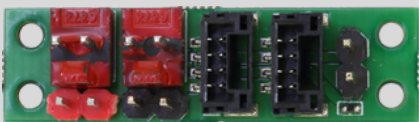
ultrasonic distance
sensor - HC-SR04



motor driver -
mrm-mot3x3.6can



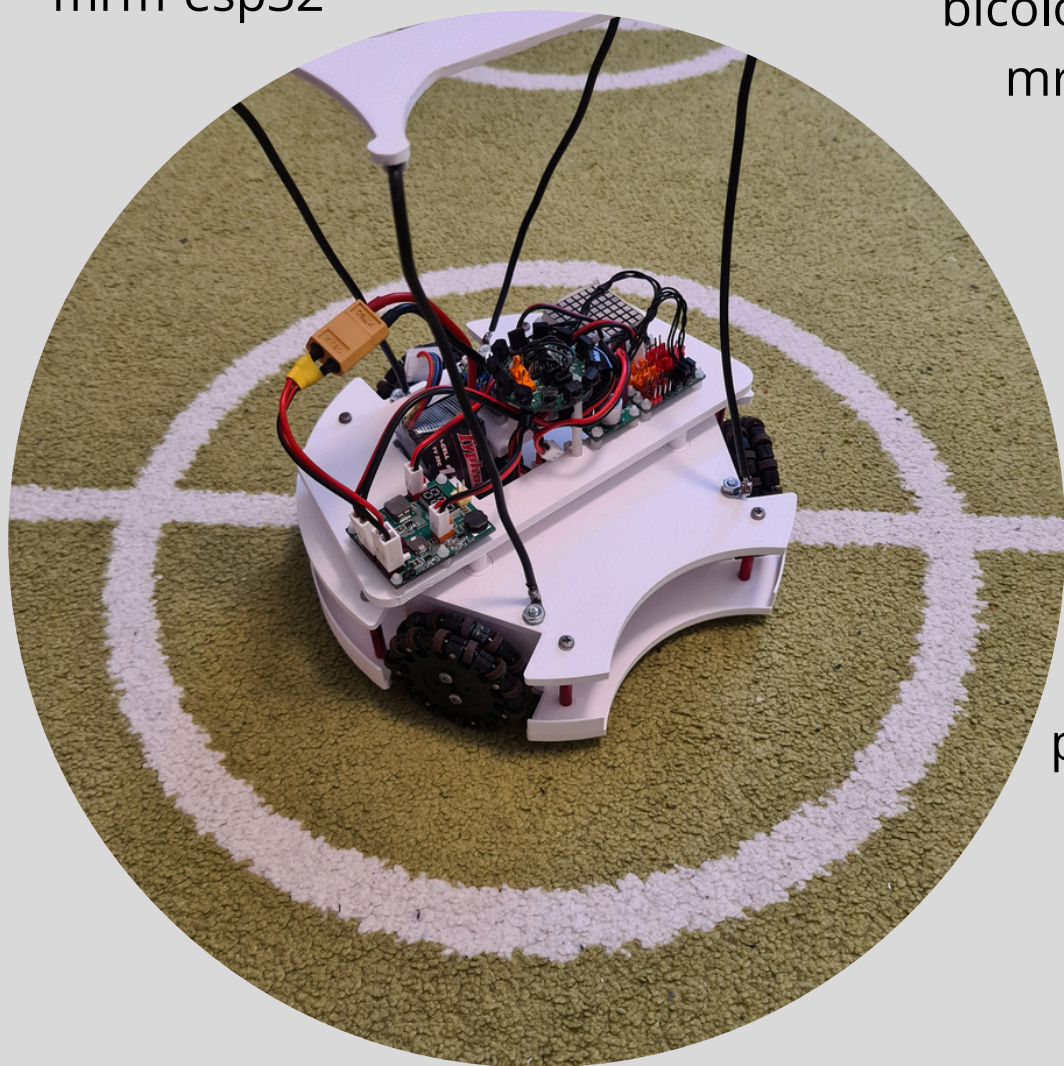
motor - RobotZone
planetary gear motor



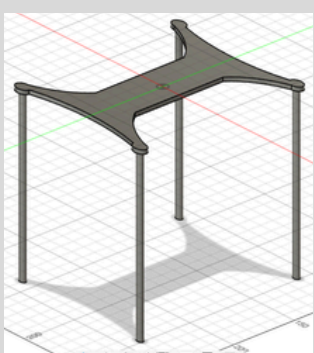
CAN Bus adapter -
mrm-can-pow



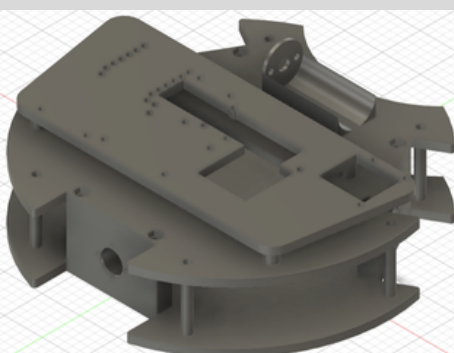
LiPo battery



wheels



handle



chassis